ANNOTATION AND TAXONOMY EVALUATION GUIDE

Improved TaxoGen - Human Annotation Guide

Annotation website: https://gannhan.ituhl.org/

1. Evaluation Purpose

This document explains how human annotators use the web-based annotation system to evaluate the quality of the taxonomy generated by Improved TaxoGen. The evaluation follows the original TaxoGen protocol and focuses on two main aspects: the correctness of parent-child relations and the semantic coherence of terms within the same taxonomy node.

Item	**Description**
Annotation website	https://gannhan.ituhl.org/
Evaluation source code	https://github.com/HoangTaiVan/ Improved_TaxoGen/tree/master/evaluation
Exported results	CSV files for each annotator, used to compute accuracy and Fleiss' Kappa

2. Annotation Tasks

The web system supports two annotation tasks:

- Intrusions: the annotator selects one term that does not fit among six displayed terms. This task is used to evaluate Term Coherence.
- Subdomains: the annotator reads the child and parent term groups, then selects Yes or No to decide whether the child is a valid subtopic of the parent. This task is used to evaluate Relation Accuracy.

3. Web Annotation Interface

Annotators access the annotation website, sign in with the assigned account, and choose one of the two annotation blocks on the home page.

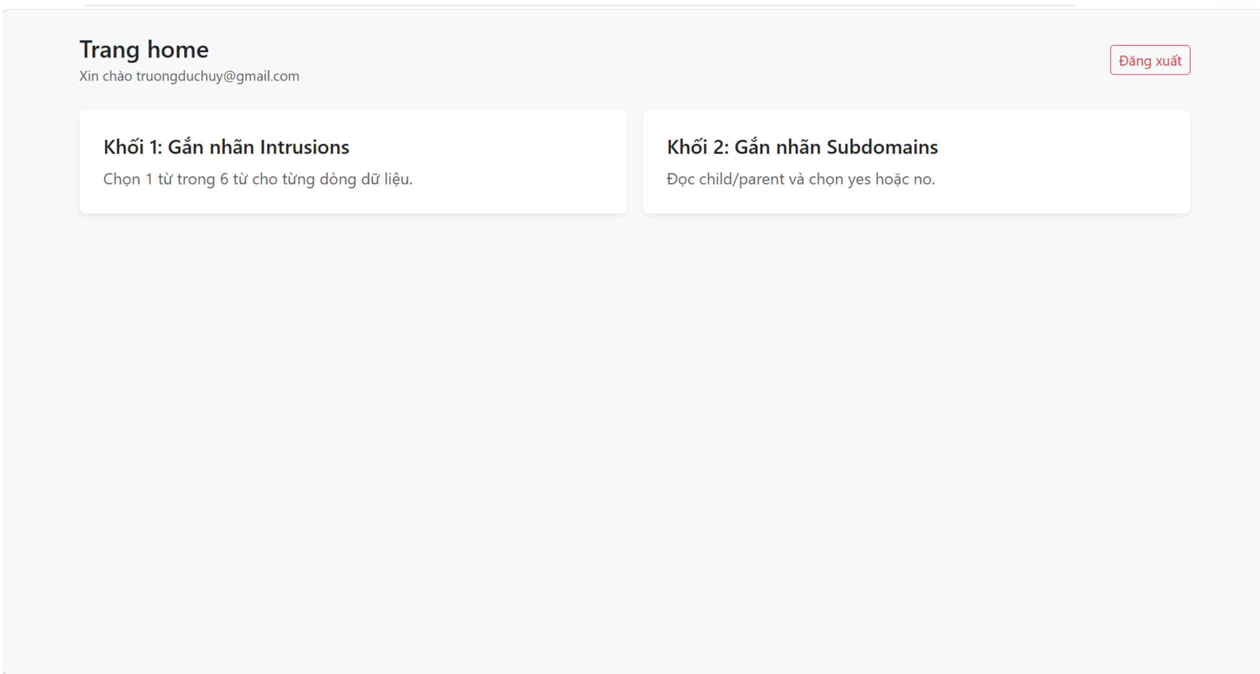

Figure 1. Home page of the annotation system, including the Intrusions and Subdomains blocks.

3.1. Topic Intrusion Annotation

In the Topic Intrusion task, each evaluation case contains six words or phrases. Five terms belong to the same topic, while one term is an intruder. The annotator selects exactly one term that is the least semantically related to the remaining terms, and then clicks Next to continue.

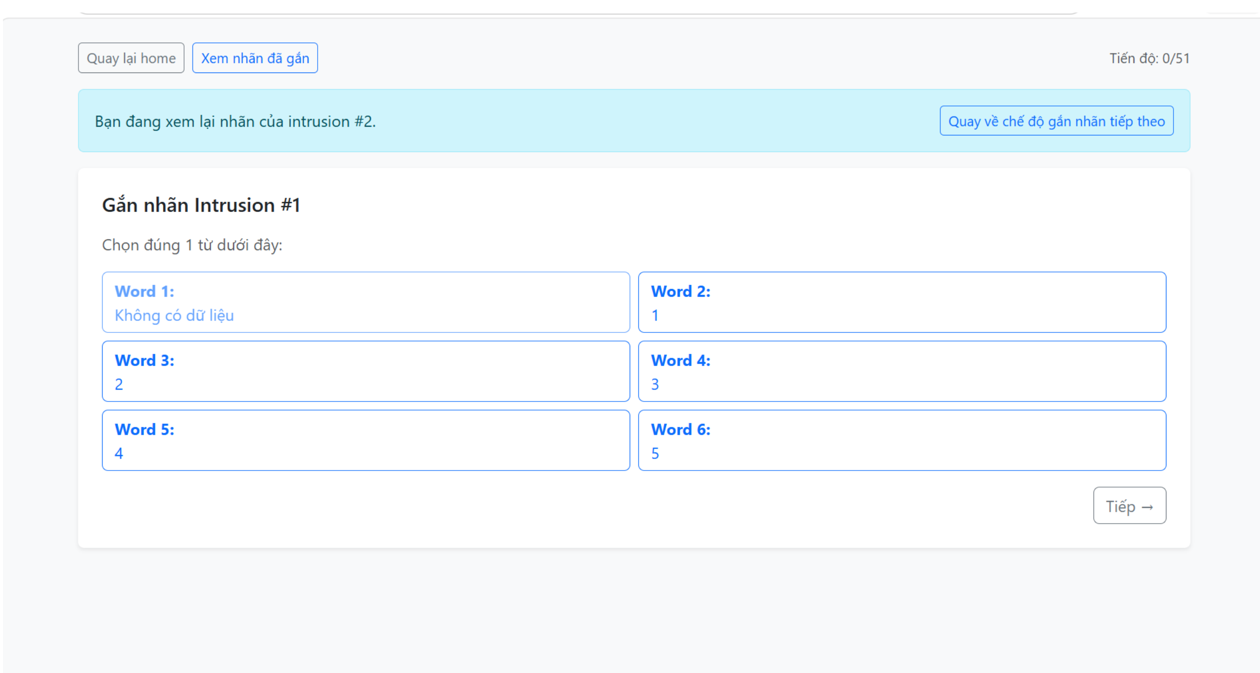

Figure 2. Intrusion annotation interface: selecting one term that does not belong to the same topic.

3.2. Parent-Child Relationship / Subdomain Annotation

In the Subdomain task, each evaluation case displays a group of terms representing the child topic and another group of terms representing the parent topic. The annotator selects Yes if the child is a valid subtopic of the parent, and selects No if the relation is incorrect, unclear, or too ambiguous.

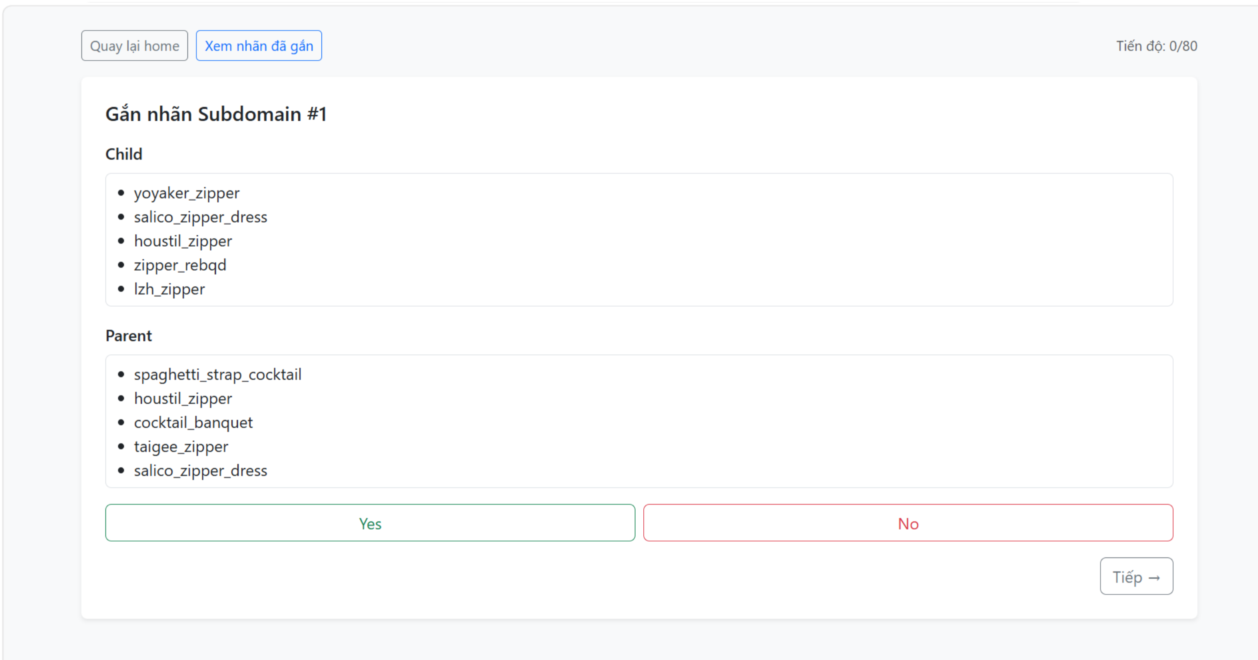

Figure 3. Subdomain annotation interface: evaluating the child-parent relation with Yes or No.

3.3. Exporting Annotation Results

After annotators complete the labeling tasks, the administrator can open the admin page and export the results. The system supports exporting CSV files for individual annotators or exporting all annotation results. These CSV files are used as input for computing Topic Intrusion Accuracy, Parent-Child Relation Accuracy, and Fleiss' Kappa.

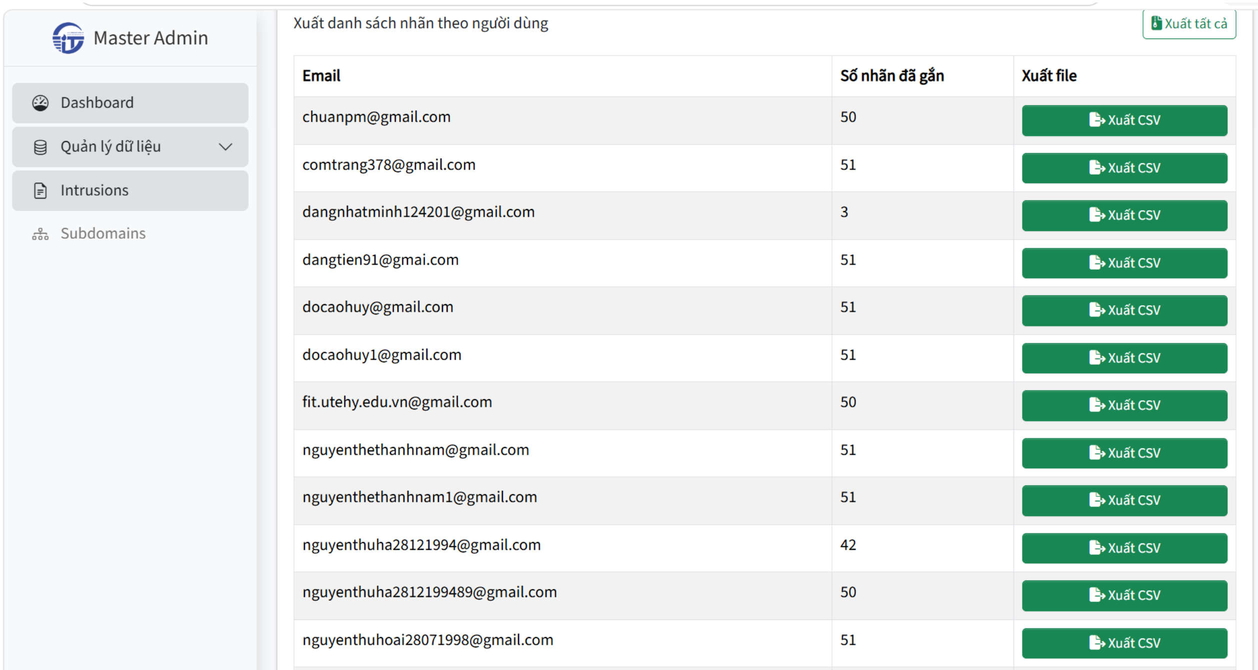

Figure 4. Admin interface: exporting annotation labels for each user as CSV files.

4. Evaluation Workflow

1. Generate evaluation files from the taxonomy using gen_eval.py.
2. Upload or place the generated evaluation files into the web annotation system.
3. Annotators sign in and complete the Intrusions and Subdomains tasks.
4. The administrator exports the annotation results for each annotator as CSV files.
5. Run run_evaluation.py to compute the evaluation metrics.

5. Commands for Sample Generation and Evaluation

Generate evaluation files:

```
py evaluation/gen_eval.py
```

After receiving the annotation CSV files from the web system, run the evaluation script:

```
py evaluation/run_evaluation.py
```

6. Computed Metrics

Metric	Meaning
Topic Intrusion Accuracy / Term Coherence	Evaluates whether the terms within the same node belong to the same semantic topic.
Parent-Child Relation Accuracy	Evaluates the correctness of hierarchical relations between parent topics and child topics.
Fleiss' Kappa	Measures inter-annotator agreement among multiple annotators beyond chance agreement.

7. Main Evaluation Files

intrusion_gold.txt
subdomain_gold.txt
intrusion_exp_0.csv
subdomain_exp_0.csv
subdomain_exp_1.csv

These files are used to generate annotation samples, store gold labels, store annotation outputs, and compute both accuracy and annotator agreement.

8. Notes for Annotators

- Read all displayed words or phrases carefully before selecting a label.
- Do not make decisions based on model names, because model names are hidden in the interface.
- For the Intrusion task, select exactly one term that is the least related to the remaining terms.
- For the Subdomain task, select Yes when the child is clearly a valid subtopic of the parent; select No when the relation is incorrect, unclear, or too ambiguous.
- Complete all assigned samples so that the annotation results can be used in the final evaluation.